

Data Analytics Capstone Topic Approval Form

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Capstone Project Name: Two-way Multivariate Analysis of Variance (MANOVA) on Marketing Campaign Dataset

Project Topic: Analysis of variance for marketing campaign data

X This project does not involve human subjects research and is exempt from WGU IRB review.

Research Question: How do language and location affect a marketing campaign's clicks, interactions, conversion rates, and ROI?

Hypothesis:

Null hypothesis

$H_{0,1}$: Language has no significant effect on the combined dependent variables

$H_{0,2}$: Location has no significant effect on the combined dependent variables

$H_{0,3}$: There is no interaction effect between language and location on the combined dependent variables

Alternate Hypothesis

$H_{A,1}$: Language has a significant effect on the combined dependent variables if the p-value is <0.05

$H_{A,2}$: Location has a significant effect on the combined dependent variables if the p-value is <0.05

$H_{A,3}$: There is an interaction effect between language and location on the combined dependent variables if the p-value is <0.05

Context: The study's contribution to the field of Data Analytics and your Master of Science in Data Analytics (MSDA) program lies in its ability to apply advanced analytical techniques to real-world marketing data, drawing actionable insights that can guide decision-making (Statistical Consulting, 2024). This study examines how language and location affect key marketing metrics like clicks, interactions, conversion rates, and ROI, leveraging a two-way MANOVA to analyse their combined influence (Hassan, 2024). These factors were chosen based on prior research indicating their significant role in shaping user engagement and campaign success. For example, studies have shown that cultural and geographic variations can influence consumer behaviour, making these variables critical in understanding and optimizing marketing strategies (Management and Strategy Institute, n.d.). By using MANOVA, this research provides a multivariate perspective, enabling a comprehensive analysis of how these variables jointly impact marketing performance.

Data: The fictional dataset represents a marketing campaign's performance across various languages and locations, capturing metrics such as clicks, interactions, conversion rates, and ROI. Each record outlines the campaign's specifics, including target audience, channel, duration, and costs, providing a comprehensive view of factors influencing digital engagement. The dataset consists of 16 columns and 200,000 records.

The data was made available on Kaggle by user Manisha Bhattacharjee. The data has the [CC0: Public Domain](#) license, which means it is accessible by anyone and can be used with no restrictions. Data was retrieved on 12/01/2024.

<https://www.kaggle.com/datasets/manishabhattach22/marketing-campaign-performance-dataset>

Field	Data Type	Variable Type
Campaign ID	Integer (categorical)	Independent
Company	Object (categorical)	Independent
Campaign Type	Object (categorical)	Independent
Target Audience	Object (categorical)	Independent
Duration (Days)	Integer (continuous)	Independent
Channel Used	Object (categorical)	Independent

Conversion Rate	Float (continuous)	Independent
Acquisition Cost	Float (continuous)	Independent
ROI	Float (continuous)	Dependent
Location	Object (categorical)	Independent
Language	Object (categorical)	Independent
Clicks	Integer (continuous)	Independent
Impressions	Integer (continuous)	Independent
Engagement Score	Integer (continuous)	Independent
Customer Segment	Object (categorical)	Independent
Date	Object (categorical)	Independent

Limitations

This study's primary limitations include relying on a fictional dataset, which may not capture all real-world complexities of marketing campaigns. Additionally, the dataset assumes accurate and complete data collection, whereas real-world datasets often include missing or noisy data. Furthermore, the study's scope is limited to specific variables (language and location), which may overlook other influential factors like campaign launch time or user demographics. Statistical assumptions of MANOVA, such as normality and homogeneity of variance-covariance matrices, also constrain the analysis, as violations of these assumptions could impact results.

Delimitations

The study is intentionally focused on language and location as the independent variables to explore their joint effects on marketing metrics, excluding other potentially relevant factors such as seasonality or platform type. Additionally, the dependent variables are limited to clicks, interactions, conversion rates, and ROI to provide a manageable scope for analysis. The choice of a two-way MANOVA was deliberate, emphasizing multivariate relationships while excluding more complex modelling approaches like mixed-effects models.

Importance of Studying This Data

Adapting marketing strategies to language and location is crucial for optimizing consumer engagement and improving outcomes like ROI. Research has shown that localizing campaigns to fit cultural and regional preferences enhances customer connection and increases sales. For example, Coca-Cola's "Share a Coke" campaign successfully adapted its core concept to different cultural contexts, leading to a global surge in sales (Ghosal, 2023). Companies build trust and credibility with diverse audiencesby tailoring messages to cultural norms. This highlights the importance of considering language and location in marketing efforts to ensure relevant and impactful campaigns across regions.

Data Gathering: The data is available to download as a CSV file. The data contains both Quantitative and Qualitative variables. The data exploration will include looking for and handling missing values and duplicates. For cleaning, the entries will be investigated to ensure no string characters like dollar signs or other measure words (e.g. days) are present in numerical fields. The fields will be converted to the right data types. Outliers can be detected and removed using visualizations like boxplots or histograms, and the assumptions for a two-way MANOVA will be checked. Overall data sparsity is 0.0%.

Data Analytics Tools and Techniques: To conduct the two-way MANOVA, a Q-Q Plot and Shapiro-Wilk test will assess the normality of each dependent variable. At the same time, Box’s Test will check the homogeneity of variance-covariance across groups. The analysis will evaluate the main effects of the two independent variables and their interaction with the dependent variables, using Pillai's Trace as the primary multivariate test statistic (NCSS Statistical Software, n.d.). A portion of the data will be set aside for cross-validation to ensure the model's reliability and robustness. Charts, Python code, screenshots of the output tables, and a Tableau dashboard will be in the presentation layer.

Justification of Tools/Techniques: A two-way MANOVA is well-suited for this analysis as it allows for the simultaneous examination of multiple dependent variables (e.g., clicks, interactions, conversion rates, ROI) and their interaction with two independent categorical variables (e.g., language and location). This method captures the independent variables' individual and combined effects on the dependent variables while controlling for Type

I error across multiple tests. It is particularly effective for identifying patterns in datasets with interconnected variables, providing a comprehensive understanding of their relationships (Marsja, 2023).

Python is well-suited for conducting a two-way MANOVA due to its versatility, ease of use, and extensive libraries like statsmodels and scipy, which support complex statistical analyses. Compared to R or SAS, Python's flexibility and integration with other tools for preprocessing, visualization, and machine learning make it especially advantageous for end-to-end analytics workflows. Moreover, Python's open-source nature and active community provide a rich ecosystem for customizing and extending analyses, particularly useful in exploratory research (Savaram, 2023).

While SAS is often favoured in corporate settings and R is renowned for statistical computing, Python's widespread use in startups and among data scientists emphasizes its lightweight and integrative features. Research comparing these tools highlights that Python's broad adoption in deep learning and data pipelines gives it a unique edge in projects requiring scalability and automation.

Project Outcomes: The comprehensive data analysis report will provide a detailed overview of a two-way MANOVA examining the effects of language and location on marketing campaign metrics, including clicks, interactions, conversion rates, and ROI. The report will detail statistical findings such as significance levels, p-values, and effect sizes, offering a robust interpretation of results. Visualizations like heat maps and grouped bar charts will effectively illustrate the interplay between independent and dependent variables. The executive summary will distil key insights, offering actionable recommendations for optimizing campaign strategies. Reproducible Python scripts with annotations will be included to ensure transparency and facilitate replication. The report will also discuss limitations and provide research directions, paving the way for more nuanced explorations of regional and linguistic influences in marketing performance.

Projected Project End Date: 1/31/2025

Sources: Sort alphabetically by last name

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Course Instructor Signature/Date:

☒ The research is exempt from an IRB Review.

☐ An IRB approval is in place (provide proof in appendix B).

Course Instructor's Approval Status: **Approved**

Date: 12/8/2024

A handwritten signature in black ink, appearing to read "J. E. Smith", is written over the "Reviewed by:" label.

Reviewed by:

Comments: [Click here to enter text.](#)